

Supplementary Materials: The Nociceptive Stability Principle

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1 S1: Substrate Inheritance Mechanism

The main text asserts that stable systems at level L cannot be built from persistently unstable components at level $L - 1$. This section provides the formal treatment.

1.1 Hierarchical Jacobian Formulation

Consider a system at organizational level L whose dynamics arise from the interaction of components at level $L - 1$. Let $\lambda_{L-1}^{(i)}$ denote the dominant eigenvalues of the level- $(L - 1)$ linearization and $\lambda_L^{(j)}$ those of level L . Suppose the effective level- L variables X_L are smooth functions of the level- $(L - 1)$ variables X_{L-1} so that

$$\dot{X}_L = F_L(X_L) = F_L(G(X_{L-1})). \quad (1)$$

The Jacobian $J_L = \partial F_L / \partial X_L$ can be written via the chain rule as

$$J_L = \frac{\partial F_L}{\partial X_{L-1}} \frac{\partial X_{L-1}}{\partial X_L}. \quad (2)$$

Linearizing around a fixed point and diagonalizing J_{L-1} , small perturbations δX_{L-1} evolve as $\delta \dot{X}_{L-1} = J_{L-1} \delta X_{L-1}$, while δX_L follows

$$\delta \dot{X}_L = J_L \delta X_L = J_L P \delta Y_{L-1}, \quad (3)$$

where P is the basis-change matrix from X_{L-1} to eigenmodes Y_{L-1} .

In generic situations the eigenvalues of J_L are smooth functions of those of J_{L-1} ; they cannot all suddenly cross from oscillatory to monotone or vice versa without a corresponding change in substrate parameters.

1.2 Inheritance Postulate

Inheritance Postulate. *If the components at level $L - 1$ are governed by a linearization with dominant eigenvalues corresponding to $\chi_{L-1} < 1$ (underdamped oscillation) or $\chi_{L-1} \gg 1$ (overdamped rigidity), and if the higher-level dynamics do not introduce new, faster stabilizing channels, then*

the level- L system cannot maintain stable, adaptive behavior over timescales longer than those of its substrates.

Intuitively: if components at level $L - 1$ ring with low damping and long-lived oscillations, any structure built from them inherits that tendency—neural populations oscillate, pain circuits flare. Conversely, extremely overdamped components yield sluggish higher-level behavior. The only way to obtain fast, non-oscillatory higher-level dynamics is to assemble them from near-critically damped components.

1.3 Application to Pain Circuits

For nociceptive circuits, the relevant substrates include:

- **Ion channel dynamics:** NMDA receptor kinetics ($\tau \sim 100$ ms, $\chi_{\text{NMDA}} \approx 0.85$)
- **Synaptic plasticity:** LTP/LTD timescales ($\tau \sim$ minutes to hours)
- **Glial activation:** Microglial response ($\tau \sim$ hours to days)

If any substrate operates with $\chi < 0.7$, the pain circuit inherits oscillatory instability. Central sensitization can be understood as substrate-level χ degradation that propagates upward to produce macroscopic pain instability.

1.4 Timescale Hierarchy

The timescale hierarchy enforces one-directional inheritance. If a substrate mode grows on timescale τ_{L-1} , any control mechanism acting on $\tau_L \gg \tau_{L-1}$ is too slow to stabilize it. In contrast, slow modes naturally average over fast stable substrates. The near-critical bound $\chi \approx 1$ therefore propagates upward from the fastest relevant layer.

This explains why top-down modulation (e.g., cognitive strategies, descending inhibition) can *temporarily* compensate for substrate instability but cannot sustain indefinite correction: continuous corrective effort depletes resources, and any interruption allows substrate instability to reassert.

2 S2: Layered Control Architecture

Tier 3 of the χ translation stack implements adaptive control. This section details the layered architecture.

2.1 Cost Function

The objective function for χ -based pain control is:

$$J = \int_0^T \left[\sum_i \lambda_i (\chi_i(t) - \chi_{\text{target}})^2 + \sum_j \mu_j \|u_j(t)\|^2 + \nu \|\dot{u}(t)\|^2 \right] dt, \quad (4)$$

where:

- $\chi_{\text{target}} \approx 0.9$ is the adaptive setpoint
- $u_j(t)$ are control inputs (SCS amplitude, medication dose)
- λ_i, μ_j, ν are weighting coefficients balancing tracking error, control effort, and smoothness

2.2 Model Predictive Control (MPC)

MPC optimizes control inputs over a receding horizon T_h :

$$u^*(t) = \arg \min_{u(\cdot)} \int_t^{t+T_h} L(\chi(\tau), u(\tau)) d\tau \quad (5)$$

subject to:

$$\dot{\chi} = f(\chi, u) + w(t) \quad (6)$$

$$\chi_{\min} \leq \chi(t) \leq \chi_{\max} \quad (7)$$

$$u_{\min} \leq u(t) \leq u_{\max} \quad (8)$$

where $w(t)$ represents process noise and $f(\chi, u)$ is the pain dynamics model.

Implementation parameters:

- Prediction horizon: $T_h = 30$ minutes (short-term) to 24 hours (circadian)
- Control horizon: $T_c = 5$ minutes
- Sample time: $\Delta t = 1$ minute for SCS, 15 minutes for pharmacological

2.3 Reinforcement Learning Layer

For long-term adaptation beyond MPC's horizon, a reinforcement learning (RL) agent learns patient-specific dynamics:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (9)$$

where:

- State $s = [\chi(t), \dot{\chi}(t), \text{var}(\chi), \text{PCC stage}]$
- Action $a \in \{\text{adjust SCS, modify medication, trigger behavioral intervention}\}$
- Reward $r = -(\chi - 0.9)^2 - \lambda_{\text{side}} \cdot \text{side_effects}$

The RL layer updates MPC model parameters and setpoints based on accumulated experience.

2.4 $\mathcal{H}_\infty$ Robustness

To guarantee stability under model uncertainty, the controller satisfies:

$$\|T_{zw}\|_\infty = \sup_{\omega} \bar{\sigma}(T_{zw}(j\omega)) < \gamma_\infty \quad (10)$$

where T_{zw} is the transfer function from disturbances w to performance outputs z , and γ_∞ is the robustness bound.

This ensures bounded χ excursions even under worst-case perturbations (unexpected stressors, medication interactions, sensor noise).

2.5 Sliding Mode Control (SMC) for Boundary Enforcement

When χ approaches boundary values, SMC provides aggressive correction:

$$u_{\text{SMC}} = -k \cdot \text{sign}(s), \quad s = \chi - \chi_{\text{boundary}} \quad (11)$$

with k chosen to guarantee finite-time return to the adaptive window.

2.6 Controller Hierarchy

The layers interact as follows:

1. **SMC** (fastest): Boundary enforcement, prevents $\chi < 0.7$ or $\chi > 1.3$
2. **MPC** (minutes–hours): Optimal trajectory tracking toward $\chi = 0.9$
3. **RL** (days–weeks): Model adaptation, learns patient-specific parameters
4. **Physician override**: Safety limits, protocol adjustments

3 S3: Operational Indicators

Beyond the PCC staging system, several operational indicators provide real-time system health assessment.

3.1 Breach Duration (τ_B)

Time spent outside the adaptive window per unit observation period:

$$\tau_B = \frac{1}{T} \int_0^T \mathbf{1}\{\chi(t) \notin [0.8, 1.0]\} dt \quad (12)$$

Clinical interpretation:

- $\tau_B < 0.05$: Healthy, minimal boundary excursions
- $\tau_B \in [0.05, 0.20]$: Transitional, intervention indicated
- $\tau_B > 0.20$: Established instability, aggressive treatment required

3.2 Return Velocity (v_R)

Rate of χ recovery after perturbation:

$$v_R = \left| \frac{d\chi}{dt} \right|_{\chi \rightarrow 0.9} \quad (13)$$

Clinical interpretation:

- $v_R > 0.1 \text{ hr}^{-1}$: Strong regulatory capacity
- $v_R \in [0.02, 0.1] \text{ hr}^{-1}$: Adequate recovery
- $v_R < 0.02 \text{ hr}^{-1}$: Impaired recovery, substrate damage likely

3.3 Cross-Timescale Divergence (Δ_τ)

Discrepancy between χ measured at different timescales:

$$\Delta_\tau = |\chi_{\text{short}}(t) - \chi_{\text{long}}(t)| \quad (14)$$

where χ_{short} uses a 1-hour window and χ_{long} uses a 24-hour window.

Clinical interpretation:

- $\Delta_\tau < 0.05$: Consistent dynamics across timescales
- $\Delta_\tau \in [0.05, 0.15]$: Multi-scale stress, monitor closely
- $\Delta_\tau > 0.15$: Timescale decoupling, structural reorganization risk

High Δ_τ often precedes PCC Stage 5 transition—the system’s short-term and long-term behaviors are diverging.

3.4 Variance Ratio (R_σ)

Ratio of current variance to baseline:

$$R_\sigma = \frac{\text{var}[\chi(t)]_{\text{current}}}{\text{var}[\chi(t)]_{\text{baseline}}} \quad (15)$$

Clinical interpretation:

- $R_\sigma < 1.5$: Within normal fluctuation
- $R_\sigma \in [1.5, 3.0]$: PCC Stage 2 likely, prophylactic intervention window
- $R_\sigma > 3.0$: Approaching Stage 3, active treatment indicated

4 S4: API Structure for Clinical Implementation

A standardized API enables integration with clinical systems.

4.1 Core Endpoints

POST /chi/estimate

Input: { "eeg": [...], "ecap": [...], "hrv": [...], "cpm": float }
Output: { "chi": float, "confidence": float, "regime": string }

GET /pcc/stage

Input: patient_id, time_window
Output: { "stage": int, "markers": {...}, "lead_time": float }

POST /control/optimize

Input: { "current_chi": float, "target_chi": float,
 "constraints": {...} }
Output: { "scs_amplitude": float, "medication_adjustment": {...} }

GET /indicators/dashboard

Input: patient_id

Output: { "breach_duration": float, "return_velocity": float,
 "cross_timescale_divergence": float,
 "variance_ratio": float }

4.2 Event Triggers

ALERT: chi_breach

Condition: $\chi < 0.75$ OR $\chi > 1.25$ for > 30 minutes

Action: Notify clinician, escalate control intensity

ALERT: variance_spike

Condition: $R_{\sigma} > 2.0$ sustained > 24 hours

Action: Flag PCC Stage 2, recommend prophylaxis

ALERT: divergence_warning

Condition: $\Delta\tau > 0.12$ sustained > 48 hours

Action: Structural reorganization risk, consider imaging

5 S5: Falsification Matrix

The framework makes seven falsifiable predictions. Each specifies the observable, method, and outcome that would invalidate the hypothesis.

Table 1: Complete Falsification Matrix

#	Prediction	Method	Falsification Criterion
1	χ_{pain} correlates with pain intensity	Prospective cohort with continuous χ monitoring	No significant correlation ($r < 0.2$, $p > 0.05$) between χ deviation from 0.9 and VAS scores over 6 months
2	CPM magnitude $\propto$ damping coefficient γ	Cross-sectional study: CPM testing + EEG-derived γ	Correlation $r < 0.3$ between CPM% and γ estimates across $n > 100$ subjects
3	PCC Stage 2 precedes chronification by 3–6 months	Longitudinal study: acute pain patients tracked to chronification	Variance increase does not precede mean drift, or lead time < 1 month or > 12 months in $> 50\%$ of cases
4	Closed-loop SCS targeting $\chi = 0.9$ outperforms fixed-parameter SCS	RCT: χ -adaptive vs. standard closed-loop SCS	No significant difference in 50% pain reduction rates at 12 months ($p > 0.05$)
5	Theta-band ($\omega \approx 7\text{--}9$ Hz) dominates chronic pain oscillations	MEG/EEG spectral analysis in chronic pain cohort	Peak pain-related power lies outside theta band in $> 60\%$ of subjects
6	Substrate χ degradation precedes macroscopic pain instability	Multi-level recording (spinal LFP + cortical EEG) in animal models	Cortical χ instability appears <i>before</i> spinal substrate changes in $> 50\%$ of cases
7	χ -guided medication titration reduces side effects vs. symptom-guided	RCT: χ -based vs. standard opioid/SNRI titration	No reduction in side effect burden (SOWS, anticholinergic score) at equivalent efficacy

5.1 Experimental Design Notes

Prediction 1 requires continuous wearable χ estimation (HRV-based proxy acceptable for ambulatory monitoring) with daily VAS collection. Power analysis: $n = 80$ for $r = 0.3$ detection at $\alpha = 0.05$, $\beta = 0.80$.

Prediction 3 requires prospective enrollment of acute pain patients (e.g., post-surgical, acute LBP) with monthly χ variance assessment. Chronification defined as pain persisting > 3 months.

Prediction 4 is the definitive clinical trial. Requires collaboration with SCS manufacturers to implement χ -targeting algorithms on existing closed-loop platforms.

Prediction 6 is the mechanistic validation. Requires animal models with multi-site electrophysiology during pain induction protocols.

6 S6: Cross-Domain Validation

The χ framework applies identically across neural instability conditions. Figure 1 demonstrates the methodology applied to dementia differential diagnosis, validating the generality of $\chi(t)$ trajectory analysis.

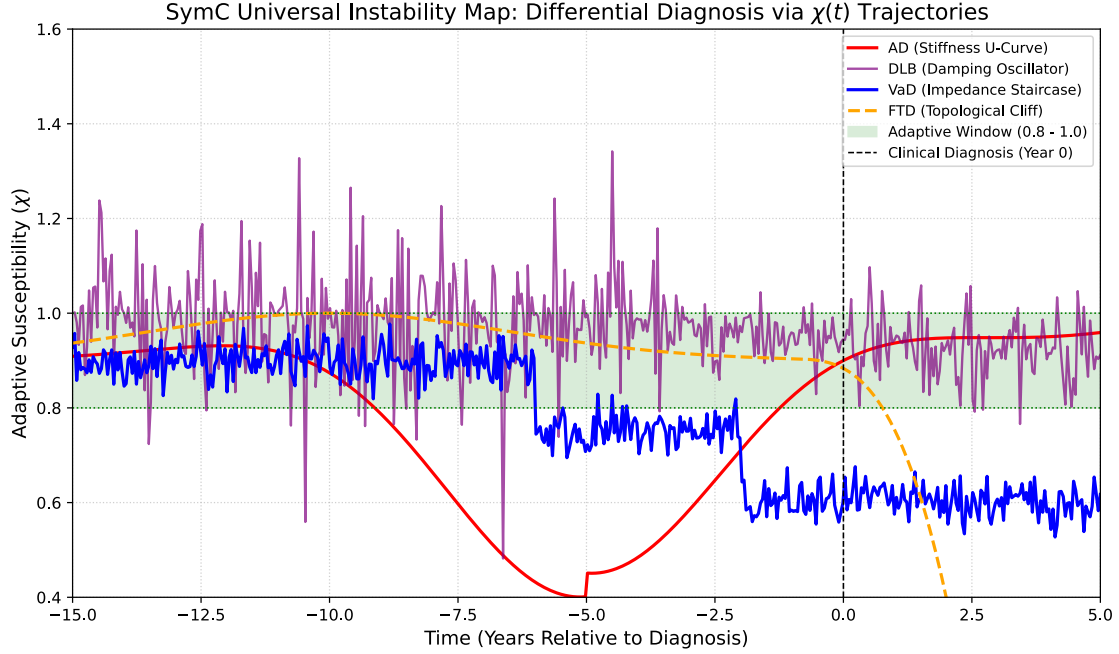


Figure 1: Cross-domain demonstration: $\chi(t)$ trajectory signatures for differential diagnosis of dementia subtypes. The same methodology that classifies pain phenotypes also distinguishes Alzheimer’s disease (stiffness U-curve: early underdamped instability transitioning to late overdamped rigidity), Dementia with Lewy Bodies (damping oscillator: high-variance fluctuations around the adaptive window), Vascular Dementia (impedance staircase: step-function drops following discrete vascular events), and Frontotemporal Dementia (topological cliff: paradoxical normality followed by sudden collapse). Each condition exhibits a characteristic dynamical fingerprint years before clinical diagnosis. The green band marks the adaptive window ($\chi \approx 0.8\text{--}1.0$); the dashed vertical line indicates clinical diagnosis (Year 0).

This cross-domain consistency validates SymC as a general stability biomarker methodology: the same χ extraction, the same PCC staging, and the same operational indicators apply whether the system is a pain circuit, a cognitive network, a cardiac pacemaker, or a crustal fault zone. The universality arises from the shared second-order operator structure governing all these systems.

References

References follow the main text.